Solutions to MP352 Problem Set 1

P.1

As we said in lecture, even though the Lorentz-FitzGerald contraction phenomenon is correct, the reason why it happens is different from what both Lorentz and FitzGerald thought. But this aside, let's figure it out anyway: if $v = 30\,\text{km}\cdot\text{s}^{-1} = 3\times10^4\,\text{m}\cdot\text{s}^{-1}$, then $v/c = 10^{-4}$. This gives $\sqrt{1-v^2/c^2} = 1 - 5\times10^{-9}$ (i.e. 0.999999995). Since the Earth's radius is $6371\,\text{km}$, its diameter is $12742\,\text{km} = 1.2742\times10^7\,\text{m}$. Thus, taking this to be L_0, we find that the contracted diameter in the direction of motion is

$$(1-5\times10^{-9})(1.2742\times10^7\,\text{m}) = 12742\,\text{km} - (5\times10^{-9}\cdot 1.2742\times10^7)\,\text{m}$$
$$= 12742\,\text{km} - 6.37\times10^{-2}\,\text{m}$$

so the Earth's diameter has contracted by $\boxed{6.37\,\text{cm}}$. Relative to the overall diameter, that's not much, but it's an amount that we can visualise.

The more modern interpretation of this result is that an observer in a frame S in which the Earth is moving at $30\,\text{km}\cdot\text{s}^{-1}$ sees its diameter in the direction of motion as $6.37\,\text{cm}$ less than an observer in the rest frame of the Earth does. No ether needs to be mentioned at all in this version of the phenomenon.

P.2

The general form of the transformation from S with coordinates (t,x,y,z) to S' which is moving at an arbitrary constant $\vec{v}$ (i.e. not necessarily in the x-direction, as we've often assumed) is

$$t' = \gamma(v)\left(t - \frac{\vec{v}\cdot\vec{r}}{c^2}\right)$$

$$\vec{r}\,' = \vec{r} + \frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v} - \gamma(v)\vec{v}t.$$

We want to confirm this does indeed give $(s')^2 = s^2$, as well as reduce to our familiar case where $\vec{v} = v\hat{e}_x$.

(a) Let's compute $\vec{r}'\cdot\vec{r}' = (x')^2+(y')^2+(z')^2$ first. To do this, recall that if $\vec{a}$ and $\vec{b}$ are two vectors, $(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b}) = \vec{a}\cdot\vec{a}+\vec{b}\cdot\vec{b} + 2\vec{a}\cdot\vec{b}$. This means that we'll have several terms in $\vec{r}'\cdot\vec{r}'$:

$$\vec{r}'\cdot\vec{r}' = \left[\vec{r} + \frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v} - \gamma(v)\vec{v}t\right]\cdot\left[\vec{r} + \frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v} - \gamma(v)\vec{v}t\right]$$

$$= \vec{r}\cdot\vec{r} + \left[\frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v}\right]\cdot\left[\frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v}\right] + \left[-\gamma(v)\vec{v}t\right]\cdot\left[-\gamma(v)\vec{v}t\right]$$

$$+2\vec{r}\cdot\left[\frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v}\right] + 2\vec{r}\cdot\left[-\gamma(\vec{v})\vec{v}t\right] + 2\left[\frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})\vec{v}\right]\cdot\left[-\gamma(v)\vec{v}t\right].$$

Since $\vec{v}\cdot\vec{v} = v^2$, this gives

$$\vec{r}'\cdot\vec{r}' = \vec{r}\cdot\vec{r} + \frac{(\gamma(v)-1)^2}{v^2}(\vec{v}\cdot\vec{r})^2 + \gamma^2(v)v^2t^2 + \frac{2(\gamma(v)-1)}{v^2}(\vec{v}\cdot\vec{r})^2$$

$$-2\gamma(\vec{v})(\vec{v}\cdot\vec{r})t - 2\gamma(v)(\gamma(v)-1)(\vec{v}\cdot\vec{r})t$$

$$= \vec{r}\cdot\vec{r} + \frac{\gamma^2(v)-1}{v^2}(\vec{v}\cdot\vec{r})^2 + \gamma^2(v)v^2t^2 - 2\gamma^2(v)(\vec{v}\cdot\vec{r})t$$

Now, if we use the fact that

$$\gamma^2(v)-1 = \frac{1}{1-v^2/c^2} - 1 = \frac{v^2/c^2}{1-v^2/c^2} = \gamma^2(v)v^2/c^2$$

then this becomes

$$\vec{r}'\cdot\vec{r}' = \vec{r}\cdot\vec{r} + \gamma^2(v)\left[v^2t^2 - 2(\vec{v}\cdot\vec{r})t + \frac{(\vec{v}\cdot\vec{r})^2}{c^2}\right]$$

Now we compute $(s')^2$:

$$(s')^2 = -c^2(t')^2 + \vec{r}'\cdot\vec{r}'$$

$$= -c^2\gamma^2(v)\left(t - \frac{\vec{v}\cdot\vec{r}}{c^2}\right)^2 + \vec{r}\cdot\vec{r} + \gamma^2(v)\left[v^2t^2 - 2(\vec{v}\cdot\vec{r})t + \frac{(\vec{v}\cdot\vec{r})^2}{c^2}\right]$$

$$= -c^2\gamma^2(v)\left[t^2 - \frac{2(\vec{v}\cdot\vec{r})}{c^2}t + \frac{(\vec{v}\cdot\vec{r})^2}{c^4}\right] + \gamma^2(v)\left[v^2t^2 - 2(\vec{v}\cdot\vec{r})t + \frac{(\vec{v}\cdot\vec{r})^2}{c^2}\right] + \vec{r}\cdot\vec{r}$$

$$= \left[-c^2\gamma^2(v) + \gamma^2(v)v^2\right]t^2 + \vec{r}\cdot\vec{r}$$

but

$$-c^2\gamma^2(v) + \gamma^2(v)v^2 = (-c^2+v^2)\gamma^2(v) = -c^2\left(1-\frac{v^2}{c^2}\right)\cdot\frac{1}{1-v^2/c^2} = -c^2$$

so

$$\boxed{(s')^2 = -c^2t^2 + \vec{r}\cdot\vec{r} = s^2}$$

and so the general transformation given does indeed leave s^2 invariant.

(b) Now to check it reduces to the correct form when $\vec{v} = v\hat{e}_x$. Since $\vec{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z$, $\vec{r}\cdot\vec{v} = vx$.

Thus,

$$t' = \gamma(v)\left(t - \frac{\vec{v}\cdot\vec{r}}{c^2}\right) = \gamma(v)\left(t - \frac{v}{c^2}x\right).$$

which is correct. Now, the last two terms in $\vec{r}\,'$ are both parallel to $\vec{v}$, and so the y- and z-cmpts of $\vec{r}\,'$ do not have these terms. This means $(\vec{r}\,')_y = (\vec{r})_y$ and $(\vec{r}\,')_z = (\vec{r})_z$, i.e. $\boxed{y' = y \text{ and } z' = z}$. Finally, the x-cmpt gives

$$
\begin{aligned}
(\vec{r}\,')_x &= (\vec{r})_x + \frac{\gamma(v)-1}{v^2}(\vec{v}\cdot\vec{r})(\vec{v})_x - \gamma(v)(\vec{v}_x)t \\
&= x + \frac{\gamma(v)-1}{v^2}(vx)v - \gamma(v)(v)t \\
&= x + (\gamma(v)-1)x - \gamma(v)vt = \boxed{\gamma(v)(x - vt)}
\end{aligned}
$$

which, since $(\vec{r}\,')_x = x'$, is correct.

P.3

This problem mostly involves manipulation of the various hyperbolic trig functions sinh, cosh and tanh.

(a) The basic eq'n for the cosh & sinh is that

$\cosh^2\chi - \sinh^2\chi = 1$

for any χ. If we divide this by $\cosh^2\chi$, this gives

$1 - \tanh^2\chi = \frac{1}{\cosh^2\chi}$.

Now, if $\cosh\chi = \gamma(v) = \frac{1}{\sqrt{1-v^2/c^2}}$, then $\frac{1}{\cosh^2\chi} = 1 - v^2/c^2$, so $1 - \tanh^2\chi = 1 - v^2/c^2 \Rightarrow v^2 = c^2\tanh^2\chi$. Taking the square root gives $\boxed{v = c\tanh\chi}$. (A caveat: technically, v could be either $c\tanh\chi$ or $-c\tanh\chi$. However, if we take the convention that v & χ are always positive – since $v = |\vec{v}| \geq 0$ – then we need to take the positive square root of $c^2\tanh^2\chi$).

Now, $\cosh^2\chi = 1 + \sinh^2\chi$ implies that $\cosh^2\chi > \sinh^2\chi$ for all values of χ. Thus, $1 > \frac{\sinh^2\chi}{\cosh^2\chi} = \tanh^2\chi$, and thus $\tanh^2\chi$ is strictly less than 1. But we've just shown, $\tanh\chi = v/c = |\vec{v}|/c$, so $\tanh^2\chi < 1 \Rightarrow \boxed{|\vec{v}| < c.}$

(b) Just to remind you of two things that you all really should know by this point in your undergraduate maths physics careers: $a\ln x = \ln(x^a)$ and $e^{\ln x} = x$. We'll need both of these for this problem.

4

The claim is that $\chi = \frac{1}{2}\ln\left(\frac{1+v/c}{1-v/c}\right)$; to show this, let's exponentiate it:

$$e^{\chi} = e^{\frac{1}{2}\ln\left(\frac{1+v/c}{1-v/c}\right)} = e^{\ln\left(\sqrt{\frac{1+v/c}{1-v/c}}\right)} = \sqrt{\frac{1+v/c}{1-v/c}}.$$

$e^{-\chi}$ is just the inverse of this, $\sqrt{\frac{1-v/c}{1+v/c}}$. Thus,

$$\cosh\chi = \frac{1}{2}(e^{\chi}+e^{-\chi})$$

$$= \frac{1}{2}\left(\sqrt{\frac{1+v/c}{1-v/c}} + \sqrt{\frac{1-v/c}{1+v/c}}\right)$$

$$= \frac{1}{2}\left(\frac{(\sqrt{1+v/c})^2}{\sqrt{1-v/c}\cdot\sqrt{1+v/c}} + \frac{(\sqrt{1-v/c})^2}{\sqrt{1+v/c}\cdot\sqrt{1-v/c}}\right)$$

$$= \frac{1}{2}\left(\frac{1+v/c}{\sqrt{1-v^2/c^2}} + \frac{1-v/c}{\sqrt{1-v^2/c^2}}\right) = \frac{1}{\sqrt{1-v^2/c^2}} = \boxed{\gamma(v)}$$

as it should.

(c) This is just putting in some actual numbers: (i) $v = 10^8 c$ means $v/c = 1/3$, so the rapidity is

$$\chi = \frac{1}{2}\ln\left(\frac{1+1/3}{1-1/3}\right) = \frac{1}{2}\ln\left(\frac{4/3}{2/3}\right) = \frac{1}{2}\ln 2 = \boxed{0.347}$$

(ii) If $\chi = 1.5$, then

$$v = c\tanh 1.5 = c\,\frac{e^{1.5}+e^{-1.5}}{e^{1.5}-e^{-1.5}} = \boxed{0.905c} = \boxed{2.72\times10^{8}\ \mathrm{m\cdot s^{-1}}}$$

(Either form is fine.)